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**Using Data Analytics to Optimise Retail Food Waste:
A Cost and Efficiency Perspective in the UK**

By

Virginia Favero

Supervised by

Dr. Soham Ghosh

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Abstract

This dissertation examines how business analytics can help UK food retailers reduce waste-related costs and improve operational efficiency, using the Co-operative Group (Co-op) as the primary case study over a ten-year monthly panel (2015-2024). Three Research Questions (RQ) are addressed. RQ1 identifies which product categories contribute most to food waste volumes and costs: combining FAO Food Price Index sub-indices with WRAP waste tonnage data supports the fact that dairy consistently accounts for the largest share of both waste and economic cost, with the financial impact of a given volume of waste rising sharply during the 2022 commodity price peak. RQ2 examines how operational scale, food price inflation, and energy costs are jointly connected to changes in waste volumes: the results show that the contraction of Co-op's store network accounts for most of the observed 45.3% reduction in waste, while lagged dairy price inflation adds modest explanatory power during the 2021-2022 period of margin pressure. RQ3 investigates whether commodity price inflation reduces waste intensity over time; an initial negative relationship largely disappears once shared time trends are accounted for, indicating limited evidence of a sustained efficiency response. Finally, a predictive analytics framework, including a relational database schema and SQL implementation, is proposed to show how a retailer with internal transactional data could implement these findings for waste forecasting. Overall, Co-op's waste reduction appears driven primarily by structural change and financial capacity as opposed to commodity price pressures alone, with implications for how retailers set targets and report progress.

Key Words: Retail Food Waste, Food Price Inflation, Waste Intensity, Diagnostic Analytics

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1. Introduction

Food waste has become an increasingly important operational, economic, and sustainability challenge for food retailers across the United Kingdom. Large quantities of food are lost across the supply chain each year, particularly at the retail and household stages, generating substantial financial losses, where food waste is estimated to cost approximately £22 billion (£25.5 billion) per year for the UK (Harvey, 2024). At the same time, this contributes to avoidable environmental impacts, such as a surplus of greenhouse gas emissions from food production and decomposition, wasted agricultural land and water resources, methane emissions from landfill, and increased pressure on energy and transport systems. Estimates from WRAP and UK Government Statistics state that 25.9 million tonnes of food waste were generated in the UK in 2023 (Gov.UK, 2024) despite ongoing reduction initiatives.

From a business perspective, food waste constitutes a direct form of organisational inefficiency and profit leakage, and does not represent a sustainability issue alone. Retailers absorb the costs of unsold inventory, disposal, refrigeration, storage, and transport, all of which increase when waste volumes rise. These pressures have intensified in recent years due to sharp increases in electricity and food prices across Europe and the UK, and energy-intensive retail operations, mainly involving refrigerated and frozen products, are especially vulnerable to fluctuations in electricity costs. Consequently, waste generated from high-perishability products carries not only a lost product value, but also a significant embedded energy cost associated with production, storage, and distribution.

The main objective of this study is to examine how business analytics techniques can support food retailers in reducing waste-related costs and improving retail efficiency within the UK. Using the Co-operative Group (Co-op) as the principal observable unit across a ten-year panel (2015-2024), the study pursues three research questions:

- **Research Question (RQ) 1:** *Which product categories contribute most to food waste volumes and associated economic costs in UK grocery retail, and how do Food Price Index-adjusted cost estimates vary across the study period 2015-2024?*
- **Research Question (RQ) 2:** *To what extent are changes in operational scale and food commodity price inflation associated with changes in food waste volumes in UK grocery retail between 2015 and 2024, and what additional role do energy cost pressures play?*

- **Research Question (RQ) 3:** *Does commodity price inflation reduce waste intensity in UK grocery retail, and does the relationship operate simultaneously or with a one-month operational lag?*

Co-op is used as the primary unit of analysis because it is the only major UK grocery retailer to have published a continuous, publicly accessible annual food waste series of sufficient length for time-series analysis. Supplementary cross-retailer evidence is drawn from a Tesco 2022-23 category-level waste breakdown and from market structure data covering twelve UK grocery retailers, with Ocado's digital retail model discussed in more detail in Chapter 5 as a forward-looking reference point.

To address RQ1, a Food Price Index-adjusted economic cost matrix is constructed at the product category level, applying the Food and Agriculture Organization's (FAO) Food Price Index (FPI) sub-indices to WRAP category tonnage estimates across the study period. This mechanism reveals how the financial burden of wasting particular categories changed as commodity prices shifted, which is invisible in volume data alone. RQ2 examines what combination of structural factors (store network size), cost environment variables (food price inflation), and energy price context can explain the pattern of Co-op's food waste reduction over time. RQ3 isolates the efficiency dimension of this relationship: holding business scale constant through a waste intensity ratio (tonnes wasted per £bn of food revenue), it tests whether commodity price inflation is linked to lower waste intensity contemporaneously or with a one-month operational lag, and assesses the robustness of that result after removing shared time trends. Finally, a predictive analytics specification grounded in the empirical findings of RQ2 and RQ3, is developed in Chapters 5 and 6 as an additional contribution to adoption as a retail operator with access to internal data.

To clarify the procedure of this investigation, Chapter 2 reviews the relevant literature, situating food waste within retail operations, embedded energy systems, supply chain frameworks, and the role of analytics in retail decision-making. Chapter 3 details the research design, data sources, variable construction, and analytical methods; Chapter 4 presents the descriptive and regression-based results. Chapter 5 discusses the findings, their managerial and policy implications, scenario-based savings estimates, and the specification of a predictive analytics environment; and Chapter 6 summarises the conclusions and recommends directions for future research.

2. Literature Review

2.1. Food Waste in Retail Operations

Food retailers operate within complex supply chains characterised by uncertain demand, short product life cycles, and significant supply chain costs, therefore overstocking, inaccurate forecasting, and product perishability often result in excess inventory that cannot be sold before expiration. As a result, retailers incur losses not only through wasted products but also through linked refrigeration, storage, transportation, and disposal costs (Gustavsson et al., 2011). These inefficiencies are more significant in periods of rising energy prices and food inflation, when already narrow retail margins come under additional pressure. Products requiring continuous refrigeration or cold-chain logistics are more vulnerable, as higher electricity prices increase the financial cost of waste generation (Gustavsson, et al., 2011). At the same time, fluctuations in food prices and consumer demand patterns make inventory planning more difficult, increasing the likelihood of overstocking and spoilage (FAO, 2024; McEvoy, 2025).

In the UK, food waste monitoring has been led by the Waste and Resources Action Programme (WRAP), which tracks volumes across household and retail stages (WRAP, 2023b; Waste Managed, 2024). WRAP estimates that the UK generated approximately 6.44 million tonnes in 2021, of which retail contributed to 234,000 tonnes, valued at £0.74 billion out of a total £21.8 billion (2021 prices) (WRAP, 2023a). Fresh produce, dairy, and meat consistently account for a disproportionate share of retail waste because of their limited shelf life and demanding storage requirements. This category profile (established and used as the baseline reference throughout the remainder of the dissertation) is what motivates the category-level focus of RQ1 and RQ3. Eurostat and FAO benchmarking show per-capita UK household food waste is slightly above the European average for perishable and processed foods, with avoidable waste concentrated primarily within the retail and household stages of the supply chain.

Despite households generating the largest proportion of avoidable food waste, parliamentary research states that retail waste remains a significant organisational issue. Common causes include overstocking, inaccurate forecasting, and broader supply chain inefficiencies, although improved stock management, forecasting systems, and food donation initiatives have reduced waste across several product categories (Malik et al., 2024). These studies also reveal that wasting high-energy products such as dairy and meat carries greater environmental and economic consequences than less energy-intensive foods. As illustrated in Figure 2.1, dairy and meat together account for more than 40% of retail food waste by volume while also requiring refrigerated storage and rapid stock turnover. Hence, they are particularly exposed to the economic pressures explored in RQ1 and RQ3.

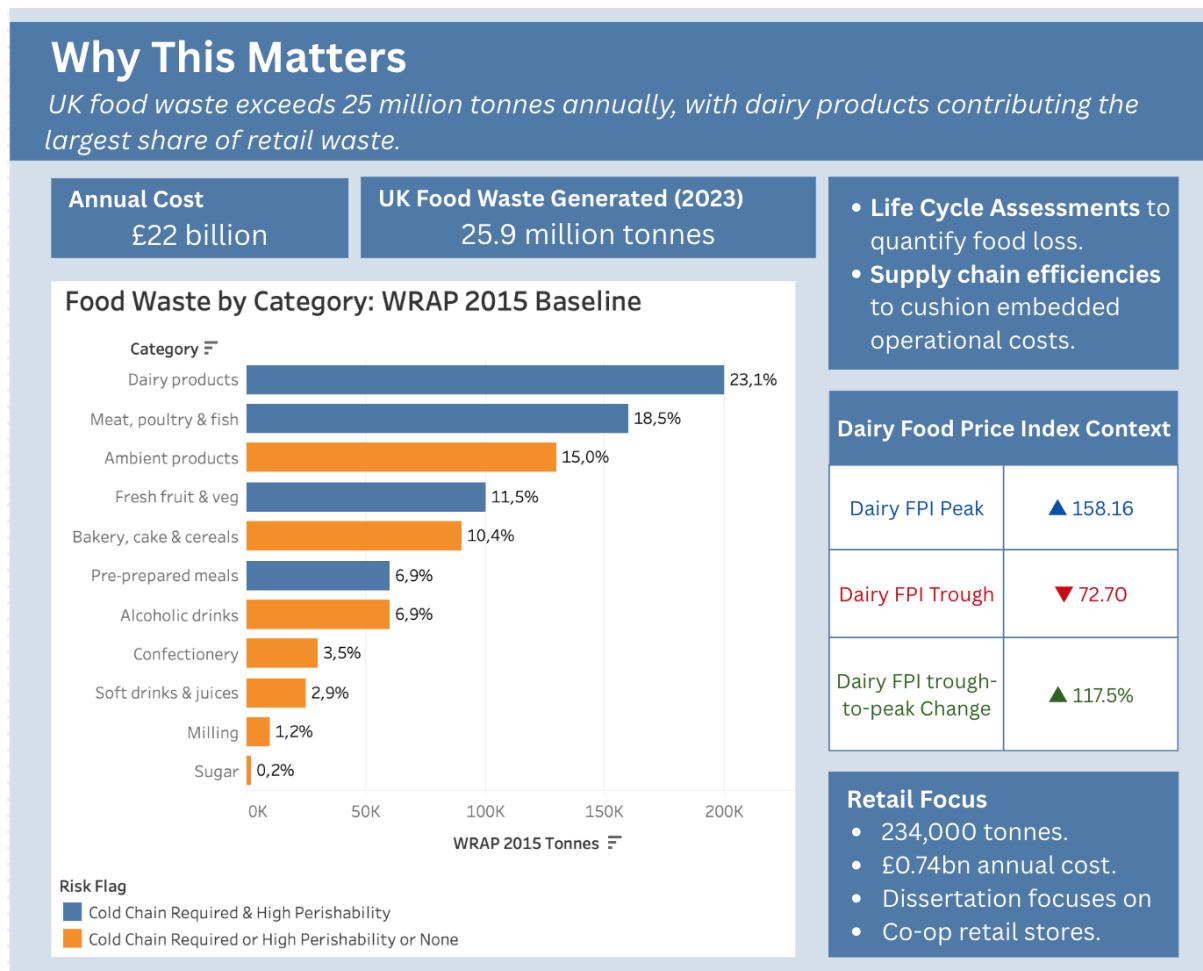


Figure 2.1: UK retail food waste context: WRAP 2015 category baseline (left), Dairy FPI trough/peak, and annual cost figures (right), motivating the category focus of RQ1 and RQ3.

Despite growing awareness, many existing waste reduction strategies remain reactive instead of being predictive. Retailers often lack integrated analytical environments linking waste generation, energy costs, and operational performance in a way that supports proactive

decision-making (O'Connor et al., 2022; WRAP, 2023b). Consequently, there is a need for analytics-based approaches that can identify the drivers of waste, estimate the economic impact of inefficiencies, and support forecasting and optimisation within retail operations.

2.2. Embedded Energy, Food Systems, and Waste Costs

The economic cost of food waste extends beyond the value of unsold products. Research undertaken by the Food and Agriculture Organization (FAO) demonstrates that wasted food also represents the loss of energy consumed across production, processing, transportation, refrigeration, and distribution stages. Globally, around one-third of all food produced is either lost or wasted, meaning considerable agricultural resources and industrial energy are consumed without delivering economic or nutritional value (FAO, 2025). High-energy-intensity categories (meat, dairy, frozen products, and heavily processed foods) carry particularly large embedded energy costs because of their reliance on energy-intensive cold chains, processing systems, and storage infrastructure. Subsequently, the true cost of food waste cannot be measured by volume alone but must also account for the embedded processes' resources associated with individual product categories. This observation motivates the category-level cost analysis in RQ1: chilled and highly perishable foods may generate disproportionately high waste-related costs despite representing smaller proportions of total physical waste volume. Life-Cycle Assessment (LCA) methodologies are increasingly used to quantify these cumulative environmental and economic impacts across supply chains (Hoehn et al., 2019).

Several studies further highlight that food waste represents a substantial hidden energy loss within modern food systems. Energy is embedded through every stage of the supply chain, including cultivation, irrigation, fertiliser production, manufacturing, packaging, refrigeration, warehousing, transport, and retail storage. When food is discarded, both the product and the accumulated energy invested in producing and distributing it are lost. FAO research postulate that inefficiencies in post-harvest storage, cooling, and transport systems significantly increase avoidable losses for perishables requiring refrigeration (Gustavsson, et al., 2011), and LCA methods are commonly applied to quantify losses in this area too (Finnveden et al., 2009; DeMenna et al., 2018). Moreover, food waste contributes an estimated 8-10% of global greenhouse gas emissions, reinforcing its importance within wider climate mitigation strategies (UK Harvest, 2024).

Within the UK, these inefficiencies create important commercial implications because supermarkets and distributors absorb not the direct cost of unsold inventory but also the associated expenditure on procurement, labour, refrigeration, logistics, storage, and energy. Rising electricity prices have amplified the financial burden of waste across chilled and frozen food categories, making waste prevention increasingly important from both a sustainability and profitability perspective. WRAP's 2024 Food Waste Reduction Roadmap reports that over 400 UK organisations now participate in structured waste-reduction initiatives, with growing adoption of measurement and analytics to identify inefficiencies across supply chains (WRAP, 2024b).

This literature informs the variable construction in Chapter 3: combining FAO FPI sub-indices with WRAP category-level waste estimates, this dissertation develops a category-specific, price-adjusted measure of retail food waste costs that provides more economically meaningful insights than waste volume alone. Dairy, meat, and fresh produce consistently emerge as the categories of highest combined perishability and energy intensity, making them the most analytically significant for any cost-focused retail waste study.

2.3. Conceptual Retail Supply Chain Framework

The conceptual food supply chain framework adopted in this study is based on the FAO food system model, which divides food-system stages into: primary production, manufacturing and processing, retail and distribution, food service, and households, each stage contributing differently to food loss, operational costs, and environmental impact. Because products move through multiple energy-intensive processes before reaching consumers, inefficiencies at any stage generate consequences across the entire system. The retail and distribution stage is positioned as the central decision-making point: retailers operate as intermediaries between upstream producers and downstream consumers, so decisions on inventory management, forecasting, replenishment, pricing, storage, and product rotation directly influence food availability and waste generation. Unlike producers or households, retailers also generate extensive operational data, making this stage well suited for business analytics applications. Predictive and descriptive analytics can thus be used to improve inventory management, reduce spoilage, optimise stock levels, and increase the efficiency of energy-intensive operations such as refrigeration and logistics.

The framework further conceptualises food waste as more than the physical disposal of products. Instead, waste represents the loss of all economic and resources invested throughout the supply chain, including procurement costs, transportation, refrigeration, labour, packaging, and storage capacity. Highly perishable products are especially vulnerable because they require continuous refrigeration and rapid inventory turnover, making them particularly sensitive to forecasting errors and operational inefficiencies. As a result, recent retail sustainability literature increasingly considers food waste to be an indicator of business performance in contrast to solely being an environmental concern. High waste-to-value ratios, repeated spoilage, and elevated disposal costs may therefore signal weaknesses in forecasting, replenishment, and demand planning. This perspective provides the conceptual foundation for the waste intensity analysis undertaken in RQ2 and RQ3.

The focus on the retail stage is further justified by data availability: retailers generate transactional, inventory, logistics, and demand data suited to forecasting models, optimisation techniques, and statistical analysis, making retail suitable for identifying patterns of overstocking, spoilage, and seasonal inefficiency. Since retailers influence supplier relationships, pricing, and consumer behaviour, interventions at this stage generate downstream benefits across the supply chain, supporting the use of Co-op's longitudinal data as the primary analytical unit in this dissertation.

2.4. Examples of Retailers: Cost Structures and Food Waste Practices

Major UK retailers, including Tesco, Lidl, Sainsbury's, and Marks & Spencer (M&S), operate complex supply chains involving procurement, transportation, warehousing, refrigeration, inventory management, and store-level distribution, where inefficiencies at the retail stage can amplify costs accumulated across the entire chain. Economic leakages occur at every stage, including at primary production, through crop spoilage, rejected produce specifications, overproduction, and weather disruptions. Instead, during manufacturing and processing, inefficiencies arise through packaging defects, contamination risks, and forecasting inaccuracies that produce excess inventory. These are intensified by energy-intensive processing and cold-chain requirements, increasing the financial impact when products are discarded before reaching consumers (UNEP, 2024).

The retail and distribution stage represents a critical managerial decision point, since retailers control inventory turnover, demand forecasting, replenishment, storage conditions,

markdown strategies, and expiry-date management. Inefficiencies at this stage (overstocking, inaccurate forecasting, poor stocking, refrigeration failures, short shelf-life products) translate into avoidable costs, on top of the embedded costs of transport, labour, warehousing, energy, and disposal. WRAP estimates food waste costs the UK food industry over £21 billion annually, concentrated in the same high-perishability, cold-chain categories identified in Chapter 2.1 (WRAP, 2022).

Several UK retailers have introduced waste-reduction initiatives combining sustainability and cost-efficiency goals (WRAP, 2024a). Tesco has implemented waste reporting, dynamic markdown systems, and improved forecasting technology, including surplus food redistribution partnerships to reduce disposal costs (Tesco, 2022). Lidl has focused on simplified supply chains, tighter inventory controls, and discounting for near-expiry products (Lidl, 2022). More broadly, discount retailers often operate with leaner structures and narrower product ranges, reducing some inventory-related waste risks compared with larger supermarket chains.

2.5. Role of Analytics in Retail Decision-Making

Existing food waste literature has largely focused on environmental sustainability, consumer behaviour, and aggregate waste volumes, with comparatively less attention given to food waste as a retail operational inefficiency and business analytics problem (Riesenegger et al., 2023; Kachhwaha and Agrawal, 2025). Research on lifecycle impacts and embedded energy has primarily emerged from environmental science perspectives, while financial studies often examine economy-wide costs instead of retailer-level decision-making. As a result, many current waste reduction approaches remain reactive, relying on markdowns, redistribution, disposal, and recycling rather than preventative predictive models (Winkler et al., 2023; Riesenegger et al., 2023).

This limitation is partly due to the complexity and fragmentation of retail data: food waste is influenced by interconnected factors, including demand fluctuations, shelf-life deterioration, promotions, seasonality, supply disruptions, and energy consumption. However, these variables are commonly managed through separate inventory, point-of-sale, refrigeration, and waste-reporting systems, preventing retailers from fully linking waste outcomes with business performance and resource use. Because of this, food waste is often treated as a

secondary operational issue and less as an optimisation opportunity (Riesenegger et al., 2023).

Current analytical approaches remain predominantly descriptive, with retailers typically relying on reporting tools and KPI dashboards to monitor waste volumes and disposal rates. Even if predictive techniques such as demand forecasting, spoilage prediction, and replenishment optimisation are increasingly discussed, more advanced methods including machine learning forecasting, dynamic pricing, and prescriptive inventory optimisation remain limited in practical adoption (Kachhwaha and Agrawal, 2025). Implementation is constrained by technological investment requirements, the need for integrated datasets, and shortages of analytical expertise. Additionally, food retail is inherently difficult to model due to perishability and demand volatility, with inflation, changing consumer behaviour, weather conditions, and energy prices affecting forecasting accuracy and increasing spoilage risks (Gustavsson et al., 2011).

A further limitation is the reliance on either aggregated national statistics or inaccessible firm-level datasets, restricting the development of transparent and scalable analytics approaches within the UK retail environment. These gaps directly inform this dissertation: RQ1 addresses limited time-varying economic estimates through category-level waste intensity and FPI-adjusted costs; RQ2 and RQ3 examine how organisational scale, inflation, and energy costs influence waste outcomes over time; and Chapter 5 develops a predictive analytics architecture assessing what public data can support and what additional internal datasets retailers require for implementation.

Overall, the literature illustrates that retail food waste represents both a sustainability challenge and a source of operational inefficiency linked to resource loss, forecasting limitations, and increasing cost pressures. While environmental impacts and waste generation patterns have been extensively examined, the application of business analytics to retail food waste management, particularly in the UK, remains underexplored. Existing research highlights gaps relating to category-level waste intensity, process drivers of waste, waste as a performance indicator, and predictive analytics applications, providing the foundation for the research questions examined in this dissertation.

6. Summary and Conclusions

6.1. Conclusions on RQ1

The first research question examined which product categories contribute most to food waste volumes and costs in UK grocery retail, and how those costs changed between 2015 and 2024 after adjusting for food price inflation. Over the study period, dairy consistently generated the highest waste volumes and costs, with meat ranking second. This pattern was also reflected in Tesco's 2022-23 food waste data, providing independent support for the category profile used in this analysis.

The principal contribution of this research, however, is not simply identifying the largest waste categories but demonstrating that waste volume and waste cost should not be treated as equivalent measures of performance. Applying FAO Food Price Index data to fixed WRAP waste volumes showed that the same quantity of dairy waste cost approximately 70-80% more at the 2022 commodity price peak than in 2015. Across all eleven product categories, the estimated monthly cost of food waste increased from under £950 million to more than £1.49 billion before easing to around £1.2 billion by 2024 as commodity prices moderated.

These findings show that fluctuations in commodity prices can substantially alter the financial impact of food waste even when the quantity discarded remains unchanged, and as a result, reporting waste solely in tonnes risks understating its economic significance during periods of inflation. This study thus provides an implementable system for assessing retailers' exposure to commodity price changes while holding waste composition constant.

6.2. Conclusions on RQ2

The second research question examined how operational scale, food price inflation and wider economic constraints influenced changes in Co-op's food waste. The results consistently showed that store count was the strongest predictor of waste, explaining most of the variation in waste levels ($R^2 = 0.773$). This implies that the contraction of Co-op's store network was the primary driver of the company's 45.3% reduction in food waste over the study period. Introducing the lagged Dairy FPI improved the explanatory power of the model ($R^2 = 0.832$), demonstrating that commodity prices also influenced waste patterns. However, the change in the coefficient's direction after controlling for store count illustrates that this relationship is more complex than implied by the initial bivariate models. In fact, commodity prices

appeared to have shaped the economic environment in which business decisions were made, as opposed to acting as a direct driver of waste.

This interpretation is reinforced by the sharp decline in Co-op's operating profit margin between 2021 and 2022, when commodity and electricity prices reached their highest levels. The largest reductions in food waste occurred only after profit margins had begun to recover, indicating that the ability to implement operational gains depends on available financial capacity and investment flexibility. Overall, the findings denote that structural change was the dominant driver of waste reduction, while commodity price inflation primarily altered the financial consequences of waste and influenced the timing of commercial responses.

6.3. Conclusions on RQ3

The third research question investigated whether commodity price inflation was associated with improvements in operational efficiency, both immediately and after a short delay. The initial regression models identified significant negative effects between commodity prices and waste intensity, reflecting that rising input costs may encourage more efficient inventory management. Nevertheless, this interpretation became less convincing once the shared long-term trends were removed, with the explanatory power of the detrended model falling to $R^2 = 0.073$. Further robustness testing showed that this did not imply the absence of a relationship. Instead, the interaction analysis revealed that the association between commodity prices and food waste was concentrated during the January 2021 to December 2023 inflationary period, when sustained cost pressures were greatest. Outside this period, no statistically significant effect was detected.

Taken together, these results reveal that commodity price inflation is not a stable or continuous driver of productivity, but its influence appears to be regime-dependent, becoming important only during periods of exceptional economic pressure. As a result, progress in food waste performance should be interpreted as outcomes shaped by the wider supply chain and financial environment in which retailers make decisions, instead of an automatic response to rising prices.

6.4. Overall Conclusions

Altogether, the findings show that food waste cannot be understood through a single measure or explained by a single driver. Instead, physical waste volumes, the financial cost of waste and operational optimisation each respond to different factors and should be assessed separately. While Co-op's reduction in food waste between 2015 and 2024 was driven primarily by structural changes, particularly the contraction of its store network, commodity price inflation substantially increased the economic cost of waste without consistently reducing waste intensity. The influence of commodity prices became apparent only during the exceptional cost fluctuations of 2021-2023, highlighting the importance of wider economic conditions in shaping business responses.

These findings illustrate the value of combining retailer organisational data with publicly available commodity-price, energy and market indicators within a business analytics environment. Doing so makes it possible to distinguish long-term structural changes from short-term market disruptions, providing a more nuanced understanding of the factors influencing food waste. From a managerial standpoint, this suggests that retailers and policymakers should complement traditional tonnage-based reporting with normalised performance measures, such as waste per store or per unit of revenue, along with category-specific targets that reflect the differing financial value of wasted products. Maintaining investment in waste reduction during periods of financial pressure is equally important, as constrained margins may delay organisational progress even when the economic cost of waste is increasing.

More broadly, this study demonstrates that publicly available data can provide informative insights into retailer performance, even where access to internal data is limited. The SQL schema proposed in Chapter 5.3 demonstrates how the empirical results can be translated and extended into a business decision-support framework, allowing future research to test the mechanisms identified in this dissertation and linking the analytical evidence developed throughout with practical retailer applications.